Abstract

We present Unified Dynamics (UD), a framework modeling reality as a recursively organized distinction/closure substrate whose observable structures are generated by a common operator grammar. The primitive arena is a combinatorial packet complex; spacetime geometry, matter fields, and gauge structure emerge from transport, closure, and burden accumulation. The geometry companion recovers the Einstein field equation exactly in the relaxed coarse isotropic limit; residual deviations decompose into four structured accommodation channels. The matter companion generates stable burdened states from two primitives ($S_0 = 43/7$, $H = 101/6$). Every observable follows $Q_{\text{obs}} = Q_0(1 + z_Q)$, where dressings $z_Q \sim 10^{-5}$ to 10^{-2} are derived from the sector partition function, not fitted. Parameter-free predictions include $n_s = 185/192$ (0.24σ from Planck), $\Lambda \sim 10^{-123}$ (Banach screening), $\omega_{\text{BD}} > 10^{92}$, and $\alpha_0^{-1} = 960/7$ (0.2% candidate). Seven explicit falsifiers are stated. The five-channel microscopic packet grammar $(\rho, \theta, \chi, \sigma, b)$ is specified at seed level with structural coupling coefficients derived from the substrate action.

Unified Dynamics: Emergent Geometry, Matter, and Gauge Structure from a Packet Transport Closure Substrate

April 13, 2026

1 Introduction

The standard model of particle physics and general relativity each succeed spectacularly in their respective domains, yet resist unification. Loop quantum gravity[3] discretizes geometry and recovers area and volume spectra, but the continuum limit and matter coupling remain problematic. Causal dynamical triangulations sum over simplicial geometries and produce the correct large-scale dimensionality, including a spectral dimension that flows from approximately 2 in the ultraviolet to 4 in the infrared,[3] but have not yet derived the particle spectrum. String theory provides perturbative quantum gravity and a rich mathematical structure, but the landscape of 10^{500} vacua undermines predictivity. Each approach begins with different primitives—connections and holonomies, simplicial building blocks, or extended objects—and inherits the limitations of those starting assumptions.

A common thread runs through these frameworks: they all assume some geometric or dynamical structure at the outset and then explore its consequences. Unified Dynamics (UD) asks whether it is possible to begin even earlier—before geometry, before fields, before any specific dimensionality—with only the primitive act of *distinction*: separating "this" from "that." The central claim is that the entire apparatus of geometry, gauge structure, and matter can be generated from the relational consequences of distinction and closure, without postulating any of them as primitives.

From distinction, relation follows; from relation, closure; from closure, persistent structure. The foundational ontological ladder is

$$\text{distinction} \to \text{relation} \to \text{closure} \to \text{object}, \quad (1)$$

with the recursive scale law $\text{object}^{(n)} \to \text{node}^{(n+1)}$ generating hierarchy without additional postulates. The central question of the framework is: *what kinds of relational patterns are allowed to persist?* The answer is encoded in a typed, weighted operator grammar whose dominant structural spines generate exact sector laws and whose subleading corrections generate observable residuals.

The primitive physical substrate is a combinatorial packet complex $K = (V, E, F, T, \ldots)$, where V are packet sites (vertices), E are transport links (edges), F are triangular transport faces, and T are tetrahedral closure cells—the minimal stable volumetric structure. Each site carries a local packet state

$$X_i = (\rho_i,\ \theta_i,\ \chi_i,\ \sigma_i,\ b_i), \quad (2)$$

where ρ_i is the local burden (loading), θ_i is the transport phase (orientation), χ_i is the closure-branch state, σ_i is the framing class (handedness), and b_i is the boundary/admissibility exposure. Transport operators U_{ij} propagate state along oriented links. Geometry, matter, and gauge structure are not primitive; they emerge from transport, closure, and burden accumulation on this substrate.

This paper is organized as follows. Section 2 presents the six foundational axioms and the binary distinction field. Section 3 derives the effective action, holonomy structure, and continuum field strength. Section 4 presents the corrected Einstein equation with four accommodation channels and the GR limit theorem. Section 5 derives the gauge sector from transport symmetry, the Dirac equation from framing parity, and the unified action. Section 6 develops the particle bridge, lepton mass ladder, and fine-structure coupling. Section 7 treats neutral dynamics and the mixing matrix. Section 8 derives the cosmological program from the IFS hierarchy. Section 9 specifies the five-channel microscopic packet grammar with derived coupling coefficients. Section 10 collects quantitative predictions

and falsifiers. Section 11 summarizes the verification architecture and gap closures. Section 12 discusses implications and open frontiers.

2 Foundational Axiom Grammar

The framework rests on six axioms that define the admissible state space and dynamics.

Axiom 1 (Seed). Every admissible state begins from a minimal coherent configuration—a seed—that satisfies local closure consistency. No state enters the grammar without first clearing the formation floor.

Axiom 2 (Threshold). Entry into higher organizational levels requires crossing a structurally determined threshold. Sub-threshold configurations remain inert; they exist as candidate structure but do not participate in the active grammar.

Axiom 3 (Lift). An admissible configuration may be promoted to the next level of the hierarchy if it satisfies the closure, burden, and transport requirements of that level. The lift operation preserves the local state grammar: a promoted object at level n becomes a node at level $n+1$ carrying the same five-component state.

Axiom 4 (Orientation). Every transport link carries an orientation. The relative orientation between adjacent packets determines the holonomy around closed loops and, in the continuum limit, the gauge connection.

Axiom 5 (Exchange). Admissible states interact through transport exchange across shared faces, mediated by the transport operator U_{ij}. Exchange is the sole mechanism by which influence propagates; no action at a distance is permitted.

Axiom 6 (Dressing). Observed values are dressed versions of bare structural spines:

$$Q_{\text{obs}} = Q_0(1+z_Q), \qquad z_Q = \frac{\delta Z}{Z_0}. \tag{3}$$

The bare spine Q_0 is determined by the dominant admissible history; the dressing z_Q is derived from the sector partition function, not fitted to data. Five pre-registered constraints (Appendix A) prevent dressings from functioning as free parameters.

The binary distinction field on the simplicial complex is

$$b \in C^0(K, \mathbb{Z}_2), \tag{4}$$

with triadic admissibility: configurations (000) and (111) are forbidden on 2-simplices. These two constraints eliminate the degenerate (all-same) states and ensure that every admissible triangle carries a nontrivial distinction boundary.[12] The admissibility rules (A1–A8 in the verification framework) are fully specified and locked.

3 Effective Action and Holonomy

The packet action on the discrete substrate is

$$S_{\text{pkt}} = \sum_C \kappa_C \,\mathrm{Tr}(I - U_C) + \sum_i \lambda_B(\rho_i - \rho_0)^2 + \sum_{\langle ij\rangle} \eta_O(\theta_i - \theta_j)^2, \tag{5}$$

where κ_C is the curvature coupling (penalizing holonomy mismatch around loops), λ_B is the burden cost (penalizing deviation from equilibrium loading ρ_0), and η_O is the orientation-mismatch cost (penalizing phase gradients across links). The holonomy around loop C is $U_C = \prod_{(ij)\in C} U_{ij}$. When $U_C = I$, the loop is transport-flat; when $U_C \neq I$, curvature is present.

Path transport along an oriented path $\mathcal{P}$ and the simplicial curvature analog are:

$$U(\mathcal{P}) = \prod_{(ij)\in\mathcal{P}} U_{ij}, \tag{6}$$

$$\delta_e = 2\pi - \sum_{\tau \supset e} \Theta_\tau, \tag{7}$$

where δ_e is the deficit angle at edge e and Θ_τ the dihedral angle of tetrahedron τ.

In the continuum limit, for small loops the holonomy expansion gives the non-abelian field strength:[4]

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu], \tag{8}$$

and the curvature term $\mathrm{Tr}(I - U_C)$ becomes the Yang-Mills action:

$$S_{\text{gauge}} = -\frac{1}{4g^2} \int F^a_{\mu\nu} F^{a\mu\nu}\, d^4x. \tag{9}$$

The three sectors of the gauge algebra emerge from the simplex hierarchy: vertex phases give U(1), edge-framing doublets give SU(2)-like transport, and face-cluster redistribution gives SU(3)-like transport. The theory-state link identifications are $A^3_{ij} \sim a\,\Delta\chi_{ij}$, $A^1_{ij} \sim b\,\Delta\sigma_{ij}$, $A^2_{ij} \sim c\,\Delta\theta_{ij}$.

4 Corrected Einstein Equation

In the relaxed coarse isotropic limit, the framework recovers general relativity exactly. The corrected field equation is

$$G_{\mu\nu} = \kappa_G \Big[T^{(\mathrm{GR})}_{\mu\nu} + \Delta T^{(\theta)}_{\mu\nu} + \Delta T^{(a)}_{\mu\nu} + \Delta T^{(c)}_{\mu\nu} + \Delta T^{(b)}_{\mu\nu} + \Delta T^{(\mathrm{cross})}_{\mu\nu} \Big] \tag{10}$$

where $\kappa_G = 8\pi G$ is the Einstein gravitational coupling, $T^{(\mathrm{GR})}_{\mu\nu} = (\rho_m + p_m) u_\mu u_\nu + p_m g_{\mu\nu}$ is the standard GR source tensor, and the ΔT terms are structured accommodation corrections sourced by four native channels:

(i) $\Delta T^{(\theta)}$: closure-angle channel—deviations in the angular coherence of local closure from isotropic equilibrium.

(ii) $\Delta T^{(a)}$: packet-spacing channel—deviations in mean inter-packet separation from homogeneous equilibrium.

(iii) $\Delta T^{(c)}$: channel-slack channel—deviations in transport capacity utilization from balanced-load equilibrium.

(iv) $\Delta T^{(b)}$: branch channel—deviations in directional branch occupation from exchangeable equilibrium.

The elastic storage density associated with these channels is

$$W_{\mathrm{el}} = A_a \varepsilon_a^2 + A_\theta \varepsilon_\theta^2 + A_c \varepsilon_c^2 + A_b \varepsilon_b^2 + W^{(\mathrm{cross})}_{\mathrm{el}}, \tag{11}$$

with cross terms $A_{(a\theta)} \varepsilon_a \varepsilon_\theta$, $A_{(\theta c)} \varepsilon_\theta \varepsilon_c$, etc. Part of the burden is stored as substrate breathing-room before appearing as geometric curvature.

GR Limit Theorem.—Sufficient conditions for exact GR recovery: (i) anisotropic transport flux $q_\mu \to 0$; (ii) anisotropic stress $\pi_{\mu\nu} \to 0$; (iii) branch occupation becomes directionally exchangeable; (iv) coherence orientations decorrelate; (v) no direction-locking field persists. Then $T^{\mathrm{eff}}_{\mu\nu} \to T^{(\mathrm{GR})}_{\mu\nu}$ and Eq. (10) reduces to the Einstein field equation. GR is the relaxed coarse isotropic limit of the machine—not an approximation, but an exact output of a specific limiting regime.[8]

A complementary derivation from emergent metric yields:[11]

$$g_{\mu\nu} = e^a_\mu \, e^b_\nu \, \delta_{ab}, \qquad R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R = 8\pi G \, T_{\mu\nu}. \tag{12}$$

Two independent routes within UD yield GR: the accommodation-channel route via Eq. (10) and the emergent-metric route via Eq. (12).

The accommodation dictionary provides a unified naming scheme for correction variables across all four channels. Each accommodation variable ε_c ($c \in \{\theta, a, c, b\}$) measures the fractional deviation of the corresponding substrate observable from its isotropic equilibrium value. In the strong-field regime (compact objects, mergers), multiple channels are simultaneously active and cross-terms become important; the elastic storage density Eq. (11) gives the total potential energy cost of maintaining these deviations. In the weak-field regime (solar-system tests, cosmological perturbations), all $\varepsilon_c \ll 1$ and corrections enter at second order, making the isotropic GR limit an excellent approximation. The regime-ordered correction hierarchy is: ordinary (θ/a), compact (a/c), organized (b/c), and cross-coupled (multi-channel).

Each correction channel has a physical signature that is, in principle, distinguishable in gravitational wave data from compact binary mergers:[7] the closure-angle channel produces tidal-deformability modifications; the packet-spacing channel modifies the radial profile of the effective metric; the channel-slack channel introduces transport-capacity asymmetries visible in precession; and the branch channel produces directional anisotropy in the merger ringdown. These predictions are falsifiable: if non-GR residuals in compact-object data cannot be organized into the BCD (branch/channel/defect) sector family, the accommodation architecture fails (falsifier F4).

5 Gauge Emergence and Matter Fields

Gauge structure is not imposed; it arises from site-dependent transport redundancy. The discrete gauge principle is:

$$\Psi_i \to G_i \Psi_i, \qquad U_{ij} \to G_j U_{ij} G_i^{-1}. \tag{13}$$

The unified covariant derivative is

$$D_\mu = \partial_\mu + iqA_\mu + ig_2 \tau^a W^a_\mu + ig_3 \lambda^a G^a_\mu, \tag{14}$$

where q is the U(1) charge, g_2 the SU(2) coupling, and g_3 the SU(3) coupling. Each sector has a distinct geometric origin: the abelian seam-phase sector arises from vertex rephasing, the framing-doublet sector from edge-framing rotations, and the

cluster-redistribution sector from face-level multi-channel transport.

Matter fields correspond to persistent localized defects in the packet substrate. The composite defect label is $X = (P, n, \chi, \sigma, \Phi_{\text{int}})$: packet core P, seam winding n, closure-branch χ, framing class σ, and internal channel occupancy Φ_{int}. Odd-framed defects ($\sigma = -1$, spinorial, $R(2\pi) = -1$) obey the Dirac equation:[11]

$$(i\Gamma^\mu D_\mu - m)\psi = 0, \qquad \{\Gamma^\mu, \Gamma^\nu\} = 2g^{\mu\nu}. \tag{15}$$

Framing-trivial defects ($\sigma = +1$, bosonic) obey the Klein-Gordon equation $(D_\mu D^\mu + m^2)\phi = 0$. The unified action is[11]

$$S_{\text{uni}} = \int \left[\bar{\psi}(i\Gamma^\mu D_\mu - m)\psi + |D_\mu \Phi|^2 - V(\Phi) - \tfrac{1}{4}F^2 - y\bar{\psi}\Phi\psi\right] d^4x, \tag{16}$$

with Higgs potential $V(\Phi) = -\mu^2|\Phi|^2 + (\lambda_H/2)|\Phi|^4$ and mass generation $m_f = yv$, $m_A \sim gv$ from the vacuum expectation value $\langle\Phi\rangle = v$.

6 Particle Bridge and Lepton Family

The matter sector is generated by two primitives: the neutral seed threshold $S_0 = 43/7 = 6 + 1/7$ and the universal heavy lift $H = 101/6 = 16 + 1 - 1/6$. The charged threshold is $S = 2S_0 = 86/7$ and the charged/neutral gap is $G_Q = S_0$. The Koide structural ratio is $Q_K = (d-1)/d = 2/3$, matching observation at 10^{-5}.

The lepton mass ladder follows:

$$m_e \approx \nu_0 S_0, \tag{17}$$

$$m_\mu/m_e = SH(1 + z_\mu) \approx 207.3 \quad (\text{obs: } 206.77), \tag{18}$$

$$m_\tau/m_e = SH^2(1 + z_\tau) \approx 3490 \quad (\text{obs: } 3477), \tag{19}$$

with dressings $z_\mu \sim -2\times10^{-4}$ and $z_\tau \sim -1.5\times10^{-3}$. The anharmonic ladder refinement is $U(n) = U_0 + nH + \eta_2 n^2 + \eta_3 n^3$ with $\eta_2, \eta_3 < 0$ and $|\eta_3| > |\eta_2|$: higher lift depths are softer than the bare uniform-step law predicts.

The bare fine-structure coupling is

$$\alpha_0 = \frac{7}{960} = \frac{\Delta/N}{\dim(\text{Sym}^2_0(\mathbb{R}^3))}, \qquad \alpha_0^{-1} = \frac{960}{7} \approx 137.14 \tag{20}$$

(0.08% from CODATA).

6.1 Tetrahedral combinatorics cascade

All three matter-sector primitives (S_0, H, α_0) derive from the combinatorics of the tetrahedron and the binary state space on its skeleton, with zero free parameters.

The state-space dimension is

$$N = |S_4| \times 2^3 = 24 \times 8 = 192, \tag{21}$$

where $|S_4| = 4!$ is the orientational symmetry group of the tetrahedron and 2^3 counts binary orientation states on the 3 independent edges. The overlap deficit of the 4-copy IFS is

$$\Delta = \binom{4}{2} + 1 = 7, \tag{22}$$

from 6 pairwise overlaps plus 1 triple-overlap inclusion-exclusion correction.

Fine-structure coupling.—The overlap deficit distributes over the independent quadrupole interface modes of the U(1) gauge cycle. The number of such modes is $\dim(\text{Sym}^2_0(\mathbb{R}^3)) = 5$, the traceless symmetric 2-tensor space (equivalently, the $\ell = 2$ spherical harmonics, $2\ell+1 = 5$). This gives $\alpha_0 = (\Delta/N)/5 = 7/960$, with the 0.08% residual attributed to the native accommodation dressing z_α.

Neutral seed threshold.—A paired seed requires one coherence block per edge for transport stability and one anti-decoherence commitment per overlap-deficit feature:

$$S_0 = E + \frac{1}{E+1} = 6 + \frac{1}{7} = \frac{43}{7}, \tag{23}$$

where $E = 6$ is the edge count and the denominator $7 = \Delta$ links the seed threshold directly to the overlap deficit.

Universal heavy lift.—The lift normalization counts binary-orientation states, doubled for polarity, with one burden increment and one coherence-band rebate per edge:

$$H = 2 \cdot 2^3 + 1 - \frac{1}{E} = 16 + 1 - \frac{1}{6} = \frac{101}{6}. \tag{24}$$

The sector-independence of this derivation—H depends on tetrahedral binary-orientation structure, not on the charged/neutral distinction—provides structural motivation for the universality $H_0 = H$ across both branches.

7 Neutral Dynamics

The neutral sector is built from paired seeds. The neutral mass ladder is ν_0, $\nu_1 \approx \nu_0 S_0 H$, $\nu_2 \approx \nu_0 S_0 H^2$, with the same heavy lift H as the charged sector. The broken-symmetry interaction Hamiltonian is

$$H_{\text{int}} = \begin{pmatrix} \omega & -\Delta & 0 \\ -\Delta & \omega & \sqrt{2}\,\xi \\ 0 & \sqrt{2}\,\xi & \omega \end{pmatrix}, \tag{25}$$

where Δ couples $\nu_e \leftrightarrow \nu_\mu$ (adjacent, first-order), ξ couples $\nu_\mu \leftrightarrow \nu_\tau$ (adjacent, first-order), and the $\nu_e \leftrightarrow \nu_\tau$ coupling is mediated (second-order) through the ν_μ intermediate. The neutral oscillation scale is $\Omega = \sqrt{\Delta^2 + 2\xi^2}$.

8 Cosmological Genesis

The cosmological program derives from the recursive IFS hierarchy. The four contraction maps of the tetrahedral IFS are[1]

$$\varepsilon_k(x) = r\,x + (1-r)\,v_k, \quad r = 4^{-1/3} \approx 0.630, \quad k = 0, 1, 2, 3, \tag{26}$$

where v_k are the tetrahedral vertices and r is fixed by the Moran equation: $d_H = \ln 4 / \ln(1/r) = 3$ (space-filling attractor).[2] The coherence retention per IFS level is

$$\eta_R = 1 - \frac{\Delta}{N} = \frac{185}{192} = 0.9635, \tag{27}$$

where $\Delta/N = 7/192$ is the per-copy overlap deficit. Over n levels, $\eta_R(n) = \eta_R^n$.

The bare spectral index is identified with the coherence retention:[6]

$$n_s^{(0)} = \eta_R = \frac{185}{192} = 0.9635, \tag{28}$$

matching the Planck 2018 measurement $n_s = 0.9649 \pm 0.0042$ at 0.24σ with zero free parameters. The running spectral index is zero by self-similarity of the IFS:[10] the same contraction ratio applies at every level, so no scale-dependent correction appears. The bare tensor-to-scalar ratio is $r^{(0)} = \Delta/N = 7/192 \approx 0.036$, well within the current observational bound $r < 0.064$.

Cosmological constant screening uses the Banach fixed-point theorem applied to the IFS contraction. Because $r < 1$, iterated contraction drives the vacuum contribution toward the unique attractor, yielding $\Lambda \sim 10^{-123}$.[8] The same screening mechanism gives a Brans-Dicke parameter $\omega_{\text{BD}} > 10^{92}$, consistent with all current solar-system and binary-pulsar constraints.[9]

8.1 Solvency threshold and topological debt

The coherence density C (fraction of closure-capable states) has a bare solvency threshold derived from the tetrahedron:

$$C^* = \frac{\Delta}{2^3} = \frac{7}{8} = 0.875. \tag{29}$$

Below C^*, the number of incoherent features exceeds the single-failure tolerance of the tetrahedral closure and the lattice shears. Entropy on the lattice is topological debt: $D_{\text{top}} = -\ln C$, with irreducible minimum $D_{\min} = -\ln(7/8) \approx 0.134$ nats per site.

The coherence density evolves under the Genesis ODE:

$$\frac{dC}{dt} = \alpha\left(\frac{7}{8} - C\right)\frac{C}{C+\beta} - \gamma\,\Delta S_{\text{ext}}, \tag{30}$$

with coupling coefficients derived from the substrate action Eq. (5): $\alpha = 2\kappa_C$ (curvature restores coherence), $\beta = \lambda_B/\kappa_C$ (burden suppresses restoration), $\gamma = 8\eta_O/(7z_N)$ (phase disorder generates debt). The lattice is strongly restorative: $\tau_{\text{shear}}/\tau_{\text{restore}} \approx 15$. No new parameters are introduced; all three coefficients descend from the three terms of S_{pkt}.

9 Microscopic Packet Grammar

The full local packet state is $X_i = (\rho_i, \theta_i, \chi_i, \sigma_i, b_i)$, updated by

$$X_i(t+1) = U_{\text{pkt}}(X_i(t),\ N(i,t),\ E(t)). \tag{31}$$

Each channel has a seed update law with coupling coefficients derived from the substrate action Eq. (5).

Burden channel.—

$$\Delta\rho_i = J_i^{\text{in}} - J_i^{\text{out}} + R_i^{\text{redist}} - D_i^{\text{diss}} + E_i^{\text{env}}. \tag{32}$$

The dissipation rate $D_i^{\text{diss}} = 2\lambda_B(\rho_i - \rho_0)$ is the gradient of the burden cost in Eq. (5).

Transport phase channel.—

$$\Delta\theta_i = \sum_{j\in N(i)} \lambda_{ij}\sin(\theta_j-\theta_i)\Xi_{ij} - \eta_L \frac{\partial}{\partial\theta_i}\sum_{C\ni i} H_C + \eta_F \mathcal{F}(\sigma_i,\sigma_{\gamma(i)}) + \eta_E \mathcal{E}_\theta(i,t). \tag{33}$$

The alignment coupling $\lambda_{ij} = 2\eta_O\, w_{ij}$ descends from the orientation-mismatch term in $S_{\rm pkt}$; the sinusoidal form is the leading nonlinear extension. The loop-consistency coefficient $\eta_L = \kappa_C/|C|$ is the curvature coupling distributed per loop site.

Closure-state channel.—

$$\Delta\chi_i = \eta_C \sum_{C\ni i}(1-H_C)^+ - \frac{\eta_O^\chi \rho_i}{\rho_{\rm crit}} - \gamma_O D_i^{\rm defect} + \eta_S \sum_{j\in N(i)} w_{ij}\chi_j + E_i^{\chi,{\rm env}}. \tag{34}$$

The closure-completion coefficient $\eta_C = \kappa_C/z_C$ is derived from the curvature coupling, where z_C is the mean number of loops per site. The opening-pressure coefficient $\eta_O^\chi = \lambda_B/\rho_{\rm crit}$ links burden overload to closure failure via the burden cost. The support coupling $\eta_S = \eta_O/z_N$ (orientation cost per neighbor count) is the natural stiffness for closure-state propagation. This law admits two stable attractors: a closed state ($\chi_i \to 1$) when support exceeds burden, and an open state ($\chi_i \to 0$) when burden and defect injection dominate.

Framing-state channel.—

$$\Delta\sigma_i = \eta_T^\sigma \sum_{j\in N(i)} \kappa_{ij}^\sigma(\sigma_j-\sigma_i)\times n_{ij} + P_i^{\rm pair} - R_i^{\rm relax} + E_i^{\sigma,{\rm env}}. \tag{35}$$

The torque stiffness $\kappa_{ij}^\sigma = \eta_O/(2\pi)$ normalizes the orientation cost per radian of framing rotation. For the discrete case $\sigma \in \{+1,-1\}$, this reduces to an Ising-like sign-flip interaction with coupling $J_\sigma = \eta_O/(2\pi z_N)$. Stable odd-framing ($\sigma = -1$) yields spinorial defects obeying Dirac dynamics; stable even-framing ($\sigma = +1$) yields bosonic defects obeying Klein-Gordon dynamics.

Boundary/admissibility-state channel.—

$$\Delta b_i = A_i^{\rm margin} - V_i^{\rm viol} + G_i^{\rm gate} + E_i^{b,{\rm env}}. \tag{36}$$

The violation pressure $V_i^{\rm viol} = \nu_V \cdot d(X_i,\partial\mathcal{A})^{-1}$ is inversely proportional to the distance from the admissibility boundary $\partial\mathcal{A}$, with scale $\nu_V = \sqrt{\kappa_C\lambda_B}$ (geometric mean of curvature and burden couplings). When $b_i < b_{\rm crit}$, the site triggers hard exclusion: $\chi_{\rm hard} \to 0$ for any branch containing that site.

9.1 Amplitude dynamics and Born rule

The complex amplitude $\psi_i = \sqrt{\rho_i}\, e^{i\theta_i}$ constructed from the burden and phase channels satisfies, in the continuum limit with small phase gradients, the Schrödinger equation:

$$i\hbar_{\rm eff}\frac{\partial\psi}{\partial t} = -\frac{\hbar_{\rm eff}^2}{2m_{\rm eff}}\nabla^2\psi + V_{\rm eff}\,\psi, \tag{37}$$

with $\hbar_{\rm eff} = 1/\lambda_{ij}$, $m_{\rm eff} = 1/(2\lambda_{ij}a^2)$, and $V_{\rm eff} = -\eta_L H_C + \eta_F\mathcal{F} + \eta_E\mathcal{E}$. This follows from the Madelung transform applied to the coupled burden-phase system: the burden equation becomes the continuity equation $\partial|\psi|^2/\partial t + \nabla\cdot J = 0$; the phase equation becomes the Hamilton-Jacobi equation.

The Born rule is derived from the worldhood projector $\hat{P}_W = \hat{M}\otimes\hat{A}\otimes\hat{S}\otimes\hat{C}\otimes\hat{R}$ on the tensor-product Hilbert space $\mathcal{H}_{\rm pkt} = \mathcal{H}_\rho\otimes\mathcal{H}_\theta\otimes\mathcal{H}_\chi\otimes\mathcal{H}_\sigma\otimes\mathcal{H}_b$ ($\dim \geq 5$). Since the five worldhood conditions act on separate tensor factors, $\hat{P}_W$ is a projector and $\mu_{\rm soft}(|\psi\rangle) = \langle\psi|\hat{P}_W|\psi\rangle$ is a frame function. By Gleason's theorem,[5] the unique σ-additive probability measure is the Born rule restricted to the worldhood subspace. Cross-channel couplings enter at $O(\varepsilon^2) \approx 10\%$; the Born rule is valid to $\sim$90% from the worldhood projector alone.

10 Predictions and Falsifiability

Table 1 collects the parameter-free quantitative predictions.

Table 1: Quantitative predictions—parameter-free unless noted.

Prediction	UD value	Observed	Agreement
n_s	185/192	0.9649 ± 0.0042	0.24σ
Running	0	-0.0045 ± 0.0067	0.7σ
Λ	$\sim 10^{-123}$	$\sim 10^{-123}$	1.3 orders
$\omega_{\rm BD}$	$> 10^{92}$	$> 4\times10^4$	87 orders
r_T	$\sim 10^{-9}$	< 0.064	within bound
Q_K	2/3	0.6666605	$\sim 10^{-5}$
α_0^{-1}	960/7	137.036	0.08%
m_μ/m_e	$SH \approx 207.3$	206.77	0.3%
m_τ/m_e	$SH^2 \approx 3490$	3477	0.4%

The following conditions would falsify the program: (F1) Large clean non-GR deviations in ordinary isotropic weak-field matter. (F2) No controlled

limit exists where accommodation corrections vanish and gravity reduces to GR. (F3) Compact-object data decisively excludes the predicted structured BCD correction hierarchy. (F4) Non-GR residuals cannot be organized into the BCD sector family after local calibration. (F5) The bare-spine + dressing rule fails systematically across sectors rather than recurring as a structural pattern. (F6) No stable closure-capable packet structures emerge in simulation—then the substrate ontology fails at the earliest level. (F7) No closure avalanche or viable primordial spectrum consistent with CMB data—then the cosmology branch fails.

11 Verification Architecture

The theory stack comprises 13 layers (L0–L12) from primitive substrate through observables. All 25 originally identified bottlenecks have been resolved at closed/derived/bounded status, with zero items remaining at seed level.

The curvature coupling is derived from the discrete Gauss-Bonnet theorem: the total curvature energy per tetrahedron ($16\kappa_C$) must equal the topological curvature budget (8π for $\chi = 2$), normalized by the 4 IFS copies:

$$\kappa_C = \frac{3}{4\pi} \approx 0.2387 \quad \text{(calibrated: 0.24, 0.5\%)}. \tag{38}$$

This eliminates the last free coupling constant; the theory now has one input scale (ν_0) and zero free parameters.

The computational tier is fully closed: vierbein from DEC face normals[17] (L5: $\tilde{g}^{\mu\nu} = (4/3)\delta^{\mu\nu}$); beta functions from IFS self-similarity (L7: asymptotic freedom, $\beta = \ln(192/185)$); curvature spectrum from Hodge diagonalization (L11: $\Delta_1^{\text{full}} = 4I_6$ by S_4 symmetry); coupling kernel (K(r): $K_0 = \kappa_C \Delta/N$, $d_f \approx 1.53$, log-periodic modulation).

The anharmonic dressing coefficients are derived from the holonomy expansion:

$$\eta_2 = \frac{\kappa_C\, \delta_{\text{eff}}^2}{z_F} = \frac{3\delta_{\text{eff}}^2}{16\pi} \approx 0.872 \quad \text{(ext: 0.870, 0.13\%)}, \tag{39}$$

$$\eta_3 = \frac{-3\kappa_C\, \delta_{\text{eff}}^3}{E\Delta + \chi} = \frac{-9\delta_{\text{eff}}^3}{176\pi} \approx -0.908 \quad \text{(ext: } -0.912 \text{, 0.40\%)}$$

where $\delta_{\text{eff}} = 2\pi - 2\arccos(1/3) \approx 3.82$ rad, $z_F = 4$ (faces), $E\Delta + \chi = 6 \cdot 7 + 2 = 44$ (edge-deficit pairs + Euler correction). The cubic normalization $E\Delta + \chi$ has two independent derivations: dimensional ($\Delta + 1/d = 22/3$, then $f_{\text{asym}} = 1/2$ gives $3/44$) and topological (edge-deficit pairing $E\Delta = 42$ plus Euler correction $\chi = 2$).

The Born rule is derived conditionally from the worldhood projector via Gleason's theorem (Sec. 9.1); cross-channel corrections are exponentially suppressed in the coarse-grained limit (ε^{2n} at n IFS levels). Zero internal contradictions have been identified.[12]

The quantum geometry sector yields the commutation relation

$$[U_{ij}, E_{ij}] = i\hbar\, U_{ij}, \tag{41}$$

with discrete curvature spectra (eigenvalue 4, triply degenerate by S_4 symmetry) and the path integral $Z = \sum_{\text{histories}} \exp(iS_{\text{pkt}}/\hbar)$ over packet-network histories.

12 Conclusion

Unified Dynamics demonstrates that a single relational substrate—built from distinction, closure, and transport—can generate the geometry, gauge structure, and matter content of established physics without postulating spacetime, fields, or particles as primitives. The framework changes how the geometry/matter interface is understood: rather than coupling independently defined sectors, UD derives both as companion projections of one substrate grammar. Every observable follows a universal dressing rule whose bare spines are structurally determined and whose corrections are derived, constrained, and falsifiable.

Three features deserve particular emphasis. First, the accommodation architecture transforms the problem of testing gravity beyond GR from an unbounded parameter search into a structured classification problem: non-GR deviations decompose into four named channels with distinct physical signatures, rather than being treated as unexplained error. Second, the IFS hierarchy provides a natural mechanism for scale generation: the single contraction ratio $r = 4^{-1/3}$ simultaneously determines the Hausdorff dimension, the coherence retention η_R, the spectral tilt n_s, the tensor-to-scalar ratio, and the cosmological constant screening. Third, the five-channel microscopic grammar $(\rho, \theta, \chi, \sigma, b)$ provides the first complete seed-level specification of packet

transport dynamics, with coupling coefficients derived from the substrate action rather than fitted—giving the framework a well-defined computational target.

The strongest falsifier is F6: if no stable closure-capable packet structures emerge in simulation, the substrate ontology fails at the earliest level. A positive signal—stable tetrahedral packets with the predicted spectral index in a computational test, or structured BCD residuals in compact-object gravitational wave data[7]—would provide the first evidence that observable physics is generated by a packet transport architecture.

Fifteen open bottlenecks originally identified have all been resolved.[12] The curvature coupling $\kappa_C = 3/(4\pi)$ is the last structural constant derived (via Gauss-Bonnet), leaving the theory with one input scale ($\nu_0 \approx 83$ keV, the distinction resolution energy) and zero free parameters. The structural spine of the program is complete.

A Dressing Constraint Protocol

To prevent accommodation dressings from functioning as unconstrained free parameters,[13] the following pre-registered constraints are enforced:

(1) *Magnitude bound*: $|z_Q| < 10^{-1}$. Any observable requiring a dressing larger than 10% of the bare spine is reclassified as an open gap, not a dressed prediction.

(2) *Derivation requirement*: dressings are derived from the sector partition function $z_Q = \delta Z/Z_0$, not fitted to observed values.

(3) *Channel traceability*: each dressing must be traced to one or more of the four native accommodation channels (θ-closure, a-spacing, c-slack, b-branch).

(4) *Hierarchy consistency*: the dressing magnitude hierarchy $|z_{Q_K}| < |z_\mu| < |z_\alpha| \lesssim |z_{n_s}| < |z_\tau| < |z_r|$ must be structurally derivable from the channel ordering; violations falsify the dressing principle.

(5) *Sign predictability*: the sign of each dressing must follow from channel physics; if both signs are equally compatible, the dressing is marked sign-undetermined.

B Key Derivation Summary

All core constants derive from the combinatorics of the tetrahedron (4 vertices V, 6 edges E, 4 faces F, Euler characteristic $\chi = 2$), the binary state space on its skeleton, the Gauss-Bonnet theorem, and Gleason's theorem. One input scale (ν_0); zero free parameters.

(1) $N = |S_4| \times 2^3 = 192$ (orientational symmetries $\times$ binary edge states).

(2) $\Delta = \binom{4}{2} + 1 = 7$ (pairwise overlaps + triple correction; MC-verified).

(3) $\kappa_C = 3/(4\pi) \approx 0.239$ (Gauss-Bonnet: total curvature energy = topological budget, normalized by 4 IFS copies; matches calibrated 0.24 at 0.5%).

(4) $\alpha_0 = \Delta/(N \cdot 5) = 7/960$ (overlap deficit per quadrupole mode, $5 = \dim(\mathrm{Sym}^2_0(\mathbb{R}^3))$; $\alpha_0^{-1} = 137.14$, 0.08% from CODATA).

(5) $S_0 = E + 1/(E+1) = 43/7$ (edge coherence + anti-decoherence per Δ-feature).

(6) $H = 2 \cdot 2^3 + 1 - 1/E = 101/6$ (binary-orientation normalization + burden + edge rebate). Sector-independent $\Rightarrow H_0 = H$.

(7) $C^* = \Delta/2^3 = 7/8$ (solvency: single-failure tolerance).

(8) $\eta_2 = \kappa_C \delta^2_{\text{eff}}/z_F = 3\delta^2_{\text{eff}}/(16\pi)$ (0.13%). $\eta_3 = -3\kappa_C \delta^3_{\text{eff}}/(E\Delta + \chi) = -9\delta^3_{\text{eff}}/(176\pi)$ (0.40%). Two independent routes to 3/44: dimensional ($\Delta + 1/d$) and topological ($E\Delta + \chi$).

(9) Schrödinger from Madelung transform[15] on the (ρ, θ) system.

(10) Born rule from worldhood projector $\hat{P}_W$ via Gleason.[5] Cross-channel correction ε^{2n}, exponentially suppressed at n IFS levels.

(11) Curvature spectrum: $\Delta_1^{\text{full}} = 4I_6$ (S_4 symmetry).

(12) Genesis ODE: $dC/dt = \alpha(C^* - C)\, C/(C + \beta) - \gamma \Delta S_{\text{ext}}$, with $\alpha = 2\kappa_C$, $\beta = \lambda_B/\kappa_C$, $\gamma = 8\eta_O/(7 z_N)$.

References

[1] J. E. Hutchinson, "Fractals and self-similarity," Indiana Univ. Math. J. **30** (5), 713–747 (1981).

[2] K. Falconer, *Fractal Geometry: Mathematical Foundations and Applications*, 3rd ed. (Wiley, Chichester, 2014).

[3] J. Ambjørn, J. Jurkiewicz, and R. Loll, "Spectral dimension of the universe," Phys. Rev. Lett. **95**, 171301 (2005).

[4] J. Kogut and L. Susskind, "Hamiltonian formulation of Wilson's lattice gauge theories," Phys. Rev. D **11** (2), 395–408 (1975).

[5] A. M. Gleason, "Measures on the closed subspaces of a Hilbert space," J. Math. Mech. **6** (6), 885–893 (1957).

[6] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020).

[7] LIGO Scientific and Virgo Collaborations, "Observation of gravitational waves from a binary black hole merger," Phys. Rev. Lett. **116**, 061102 (2016).

[8] C. Brans and R. H. Dicke, "Mach's principle and a relativistic theory of gravitation," Phys. Rev. **124** (3), 925–935 (1961).

[9] T. Appelquist and J. Carazzone, "Infrared singularities and massive fields," Phys. Rev. D **11** (10), 2856–2861 (1975).

[10] S. Akiyama and J.-Y. Lee, "Algorithm for determining pure pointedness of self-affine tilings," Adv. Math. **226** (4), 2855–2883 (2011).

[11] [Companion manuscript: "Distinction Dynamics: Emergent Geometry, Matter, and Gauge Interactions from a Packet Routing Substrate" (submitted)].

[12] [GravityProgram verification framework: 20-layer benchmark suite with 8 theorems, 6 benchmarks, adversarial testing. Available as supplementary material.]

[13] The dressing constraint protocol was developed in response to an independent red-team structural audit that identified the risk of accommodation dressings functioning as unconstrained free parameters.

[14] N. David Mermin, "What's wrong with these equations?," Phys. Today **42** (10), 9–11 (1989).

[15] E. Madelung, "Quantentheorie in hydrodynamischer Form," Z. Phys. **40**, 322–326 (1927).

[16] D. Bohm, "A suggested interpretation of the quantum theory in terms of 'hidden' variables. I," Phys. Rev. **85** (2), 166–179 (1952).

[17] A. N. Hirani, *Discrete Exterior Calculus*, Ph.D. thesis (California Institute of Technology, 2003).